EE523 – Integrated Navigation Systems

Project I

INS 3D Reduced Inertial Sensor Mechanization in Local Level Frame

Background:

In an inertial navigation system (INS), the inertial measurement unit (IMU) contains three accelerometers, which are orthogonal to each other and measure the specific force with respect to the body frame. In addition to accelerometers, the INS also contains three gyroscopes, which measure the rotation rate. This rotation rate is used to obtain the attitude and it is also used to transform the accelerometer measurements from body to local level frame. These measurements (specific force and rotation rate) are converted to position, velocity, and attitude thorough a process, known as Mechanization. In this project, given that land vehicles have speed measurements through the onboard odometer and they do not experience fast change of accelerations and their attitude angles, you will explore providing full 3D navigation solution using only one gyroscope, three accelerometers, and the vehicle odometer.

Data provided.

You will be provided with the trajectory data for the following IMUs:

1. High-end tactical (HG1700, Honeywell)
2. Low cost (automotive grade) MEMS-based (VTI IMU)

The data will contain the following:

1. Raw IMU data containing the specific force measurements from accelerometers and rotation rates from gyroscope.
2. Vehicle velocity data obtained through odometer
3. Reference data from High-end tactical grade INS which will serve as truth data for position, velocity, and attitude.

Requirements:

Implement the mechanization for 3D RISS and test the code using the raw data from the IMU. The requirements include:

1. Implement 3D mechanization of a reduced inertial sensor system with the involvement of vehicle odometer data. The code should be commented wherever appropriate for ease of understanding.
2. Test the code with the raw outputs (angular rates, specific forces and odometer measurements) from low-cost MEMS grade IMU as well as the vehicle odometer.
3. Plot the output of the mechanization (position, velocity and azimuth), the reference solution and the error between them for the whole trajectory (set of data points). Write a report that should include following:
    a. A brief overview of the errors and mechanization process.
    b. Discussion on the effects of major errors on the output of the mechanization when compared with the reference.
    c. Discussion on the behavior and accuracy of two different kind of IMUs used in the project.

d. Comment on the limitations and advantages of 3D RISS mechanization and possible improvement that could enhance the accuracy of the 3D RISS mechanization results.
e. Present the positioning performance in terms of:
    i. The 2D RMS position errors, that is the resultant of the position errors along the East and North directions.
    ii. The total 3D horizonal position error. Study and comment on the impact of the altitude errors.
    iii. The percentage of the 2D RMS position errors in terms of the distance travelled over the whole trajectory.